

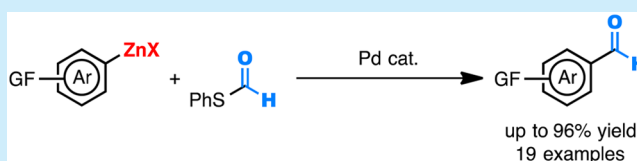
Palladium-Catalyzed Formylation of Arylzinc Reagents with S-Phenyl Thioformate

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S Supporting Information

ABSTRACT: The first example of palladium-catalyzed direct formylation of arylzinc reagents using S-phenyl thioformate is reported. The reaction proceeded under mild conditions, allowing high functional group tolerance. In addition, the developed formylation method was used to prepare deuterated and ¹³C-labeled aryl aldehydes from isotope-labeled S-phenyl thioformates. Moreover, this procedure was applied to an alkenylzinc halide, affording the corresponding enal.

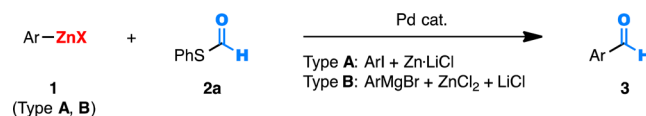


Aromatic aldehydes are versatile building blocks for various C–C bond formation reactions.¹ Among numerous aldehyde synthesis methods, the highest synthetic efficiency is achieved by the direct introduction of a formyl group into aromatic rings. Electrophilic formylation of aromatics is ideally suited for this purpose, avoiding prefunctionalization prior to formyl group incorporation.² However, the corresponding substrate scope is limited to electron-rich aromatic compounds; moreover, their functional group tolerance is low. Functionalized aldehydes can be obtained by transition-metal-catalyzed formylation,³ particularly by palladium-catalyzed reductive formylation of aryl halides with CO gas.⁴ Although this method has been applied on an industrial scale,⁴ⁱ the use of gaseous CO hampers the utilization of such procedures by synthetic chemists working in the laboratory. Recently, some reagents have been developed as C1 sources, presenting an alternative to the toxic CO gas.⁵ Aryl aldehydes have mostly been synthesized by formylation of organolithium and organomagnesium reagents with formic acid or its derivatives due to the user-friendly handling of these reagents.⁶ However, such methods suffer from low functional group tolerance because of their high nucleophilicity. Therefore, new methods for the synthesis of highly functionalized aldehydes under mild reaction conditions are still required.

Organozinc reagents are regarded as versatile C-nucleophiles due to their excellent functional group tolerance and chemoselectivity.⁷ They have exhibited enhanced synthetic value owing to their utilization in transition-metal-catalyzed cross-coupling reactions.^{8,9} For instance, the use of thioesters as coupling partners affords highly functionalized ketones under mild reaction conditions (Fukuyama coupling).⁹ Even though significant progress has been made in organozinc chemistry, direct formylation of organozinc reagents remains undeveloped because of the lack of appropriate formylating agents.¹⁰ Herein, we utilized S-phenyl thioformate as a versatile formylating agent for palladium-catalyzed formylation of arylzincs (Scheme 1). This reaction proceeded under mild conditions to afford highly

functionalized aldehydes, which are otherwise difficult to synthesize by the aforementioned conventional methods.

Scheme 1. Formylation of Arylzinc Reagents by Cross-Coupling Reaction with S-Phenyl Thioformate

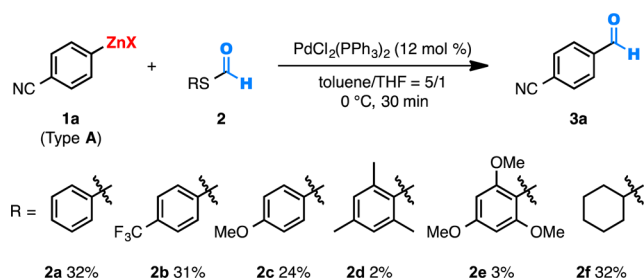


S-Aryl thioformates **2** have previously been prepared by other research groups¹¹ and utilized for O-formylation of carboxylic acids in the field of biochemistry.^{11a} However, the corresponding C-formylation of organometallic reagents has received less attention. To achieve this transformation, we explored palladium-catalyzed formylation of organozinc reagents with S-aryl thioformates. First, we screened a range of electronically and sterically diverse S-aryl thioformates in the formylation of organozinc reagent **1a** prepared from 4-iodobenzonitrile with Zn and LiCl (type A reagent). As shown in Scheme 2, treatment of **1a** with S-phenyl thioformate (**2a**) at 0 °C for 30 min in toluene/THF afforded the formylated product in 32% yield. Although the use of S-aryl thioformate **2b**, bearing an electron-withdrawing group did not result in a marked change, the presence of an electron-donating group reduced the yield (**2c**), and only traces of the product were observed for sterically demanding **2d** and **2e**. Alkylthiol-derived thioformate **2f** gave the product in 32% yield. Considering the greater availability of **2a** compared to that of **2b** and **2f**, we identified the former as an appropriate formylating reagent for this reaction and further investigated the reaction condition optimization.

After extensive experimentation (see the Supporting Information for details), formylation of **1a** with **2a** afforded

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Scheme 2. Pd-Catalyzed Formylation of Organozinc Reagent 1a with 2: Effect of Different S-Substituents^{a,b}



^aStandard conditions: 1a (0.22 mmol), 2 (0.20 mmol), $\text{PdCl}_2(\text{PPh}_3)_2$ (24 μmol), toluene (2.0 mL), THF (0.4 mL) at 0 °C for 30 min.
^bDetermined by ^1H NMR analysis of the crude reaction mixture using dibromomethane as the internal standard.

3a in 96% yield under the following optimized reaction conditions: $\text{Pd}(\text{OAc})_2$ (2.0 mol %) and $\text{P}(\text{2-furyl})_3$ (1.4 mol %) in toluene/THF = 5/1 (v/v) at 0 °C for 30 min (Table 1, entry

Table 1. Optimization of the Reaction Conditions for the Pd-Catalyzed Formylation of 1a with 2a^a

entry	variation from the standard conditions	yield ^b (%)
1	none	96
2	no $\text{Pd}(\text{OAc})_2$	trace
3	no $\text{P}(\text{2-furyl})_3$	14
4	PPh_3 , instead of $\text{P}(\text{2-furyl})_3$	24
5	PCy_3 , instead of $\text{P}(\text{2-furyl})_3$	20
6	BINAP, instead of $\text{P}(\text{2-furyl})_3$	12
7	dppf, instead of $\text{P}(\text{2-furyl})_3$	3
8	$\text{P}(\text{2-furyl})_3$ (1.0 mol %)	47
9	$\text{P}(\text{2-furyl})_3$ (2.0 mol %)	55
10	$\text{P}(\text{2-furyl})_3$ (4.0 mol %)	45
11	$\text{P}(\text{2-furyl})_3$ (8.0 mol %)	39
12	THF as a sole solvent	25
13	at 25 °C	42

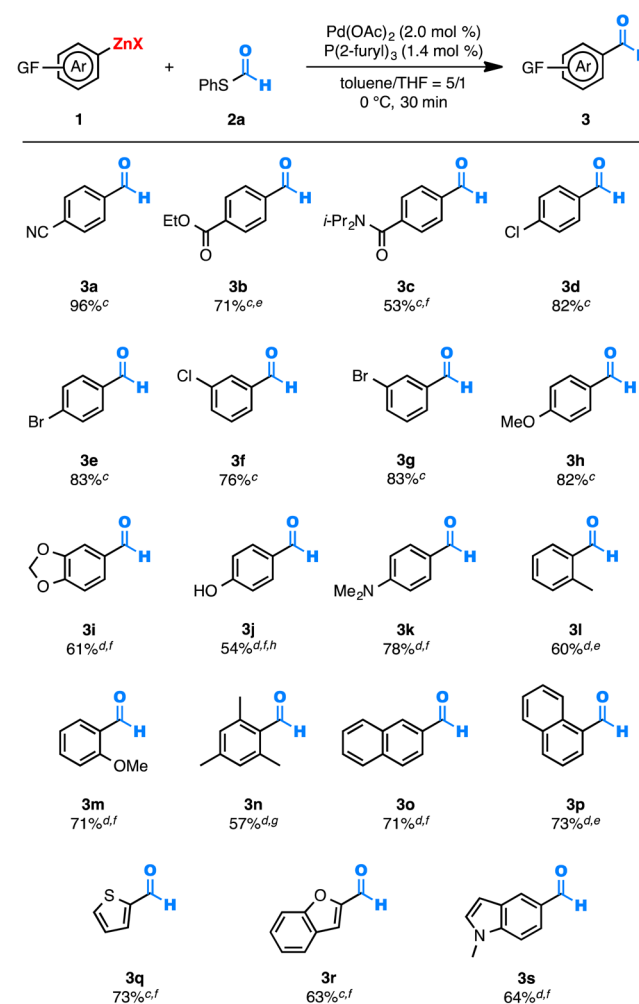
^aStandard conditions: 1a (0.22 mmol), 2a (0.20 mmol), $\text{Pd}(\text{OAc})_2$ (4.0 μmol), $\text{P}(\text{2-furyl})_3$ (2.8 μmol), toluene (2.0 mL), THF (0.4 mL) at 0 °C for 30 min. ^bDetermined by ^1H NMR analysis of the crude reaction mixture using dibromomethane as the internal standard.

1). No formylation took place in the absence of $\text{Pd}(\text{OAc})_2$ (entry 2), and a poor yield was obtained in the absence of $\text{P}(\text{2-furyl})_3$ (entry 3). As shown in entries 4–7, the use of mono- and bidentate phosphines instead of $\text{P}(\text{2-furyl})_3$ resulted in lower yields.

Notably, the amount of $\text{P}(\text{2-furyl})_3$ played a key role in this formylation, with deviations from the optimal value of 1.4 mol % causing significantly deteriorated yields (entries 1 and 8–11). This result is interesting because Fukuyama coupling usually requires more than 1 equiv of the phosphine ligand relative to palladium, although the exact reason behind the dramatic effect of the phosphine amount in the present reaction is not well understood. Conducting formylation in pure THF decreased the product yield to only 25% (entry 12). The increase of reaction temperature also reduced the yield (entry 13).

With the optimized reaction conditions in hand, we next investigated the scope of arylzinc reagents (Scheme 3), initially

Scheme 3. Substrate Scope of Arylzinc Reagents^{a,b}

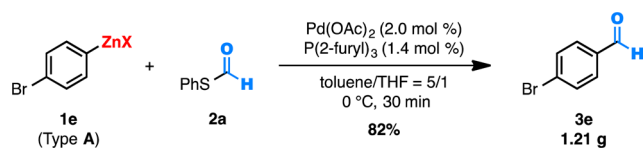


^aStandard conditions: 1 (0.22 mmol), 2a (0.20 mmol), $\text{Pd}(\text{OAc})_2$ (4.0 μmol), $\text{P}(\text{2-furyl})_3$ (2.8 μmol), toluene (2.0 mL), THF (0.4 mL) at 0 °C, for 30 min. ^bIsolated yields. ^cReagent type = A. ^dReagent type = B. ^eAt 0 °C for 2 h. ^fAt –20 °C for 15 h. ^gAt 0 °C for 15 h. ^hThe starting material was protected by a TMS group, which was removed after workup.

focusing on the synthesis of functionalized aldehydes bearing electrophilic functional groups that are not tolerated in the conventional formylation of organolithium or organomagnesium reagents. For example, arylzinc reagents containing cyano, ethoxycarbonyl, and *N,N*-diisopropylamide moieties were formylated in good yields to afford 3a–c. Arylzinc reagents with halogen functional groups underwent smooth formylation to give product 3d–g without any functional group loss. Electron-rich arylzinc reagents (3i–p,s) were prepared from the corresponding ArMgBr with equimolar of ZnCl_2 : type B reagent) could also be utilized in the reaction (3h–k). 4-Hydroxybenzaldehyde (3j) was synthesized from a TMS-protected precursor, and deprotection was performed after quenching with aqueous NH_4Cl . Substituents in the 2-position did not cause steric inhibition, furnishing the desired products in moderate to good yields (3l–n). Polyaromatic and heteroaromatic aldehydes could also be synthesized by

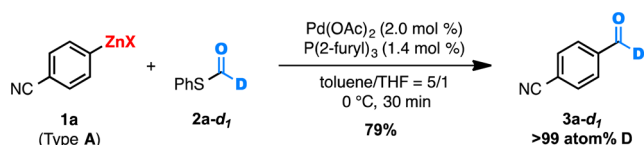
formylation (**3o-s**). The result of a gram-scale synthesis of **3e** (1.21 g, 82%) showed the practicality of the present reaction (Scheme 4).

Scheme 4. Gram-Scale Synthesis

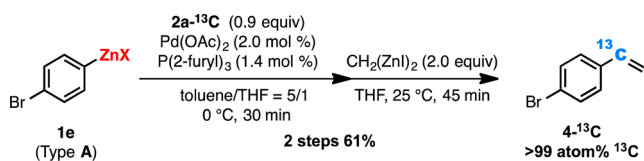


Further applications concerning the incorporation of isotope-labeled formyl groups are shown in Schemes 5 and 6. **2a-d₁** and

Scheme 5. Synthesis of the Deuterated Aldehyde



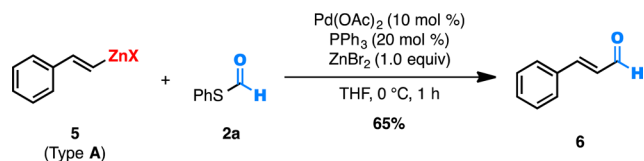
Scheme 6. Synthesis of the ¹³C-Labeled Compound



2a-¹³C were readily prepared from commercially available isotope-labeled formic acids. The reaction of **2a-d₁** with **1a** under optimized reaction conditions afforded **3a-d₁** (99 atom D %), as confirmed by ¹H NMR analysis. Moreover, isotope-labeled 4-bromostyrene **4-¹³C** was readily obtained by ¹³C-formylation of arylzinc reagent **1e** and a subsequent methylenation reaction.¹² The prepared compounds are versatile building blocks for the synthesis of isotope-labeled molecules.

As shown in Scheme 7, this method could also be applied to an alkenylzinc reagent. In the presence of Pd(OAc)₂ (10 mol

Scheme 7. Formylation of Alkenylzinc Reagent **5**



%), PPh₃ (20 mol %), and ZnBr₂ (1.0 equiv), the reaction of alkenylzinc reagent **5** with **2a** proceeded with the retention of configuration, furnishing the corresponding enal compound. Although the yield of the above reaction needs to be improved, this result suggests the possibility of further expanding the substrate scope of the present formylation.

In summary, we have demonstrated the first example of palladium-catalyzed direct formylation of arylzinc reagents with S-phenyl thioformate under mild reaction conditions to afford functionalized aldehydes bearing cyano, ester, amide, chloride, and bromide groups. The use of isotope-labeled formylating reagents allowed the introduction of isotope-labeled formyl groups. Thus, the developed method provides a safe alternative

to CO utilization, and further studies on extending its scope are currently underway in our laboratories.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.7b00447.

Detailed experimental procedures and compound characterization data (PDF)

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Notes

The authors declare no competing financial interest.

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